

# TEST PLAN

---

**Project Name:** OpenCart E-Commerce Application

**Prepared By:** Nilava Pal

**Role:** Senior QA Automation Engineer

**Version:** 1.0

**Date:** March 2025

## 1. Introduction

This document outlines the testing strategy, scope, objectives, resources, and schedule for testing the OpenCart E-Commerce web application. The purpose is to ensure the application meets functional and non-functional requirements before production release.

## 2. Objectives

- Validate core e-commerce workflows
- Ensure system stability and usability
- Detect defects early
- Verify integration between UI and backend
- Ensure release readiness

## 3. Scope of Testing

### In Scope:

- User Registration & Login
- Product Search & Filtering
- Product Details Page
- Add to Cart
- Checkout Process
- Payment Gateway Integration
- Order Confirmation
- Admin Panel Functionalities
- API Validation (Cart, Order, Login)
- Cross-browser testing

### Out of Scope:

- Third-party payment gateway internal logic
- Infrastructure performance benchmarking
- Penetration testing

## 4. Test Types

- Functional Testing
- Regression Testing
- Smoke Testing
- Integration Testing
- Cross Browser Testing
- API Testing (REST Assured)
- UI Automation (Selenium WebDriver)

- Usability Testing

## 5. Test Approach

- Manual Testing with requirement-based test cases
- Automation using Selenium WebDriver (Java)
- TestNG Framework with Page Object Model
- REST Assured for API validation
- Maven for build management
- Jenkins CI integration
- Extent/Allure reporting

## 6. Test Environment

- Application URL: <https://demo.opencart.com>
- OS: Windows 11
- Browsers: Chrome (latest), Firefox
- Automation Tools: Selenium 4, Java 17
- Build Tool: Maven
- CI/CD: Jenkins
- Version Control: GitHub

## 7. Entry Criteria

- Requirements finalized
- Test cases reviewed
- Test environment ready
- Build deployed in QA

## 8. Exit Criteria

- 95% test cases executed
- No Critical/High defects open
- Regression suite passed
- Test summary report prepared
- Stakeholder approval

## 9. Deliverables

- Test Plan
- Test Strategy
- Test Cases

- Automation Scripts
- Defect Report
- Test Execution Report
- Release Report
- Requirement Traceability Matrix (RTM)

## **10. Risk & Mitigation**

- Environment instability – Maintain backup QA environment
- Requirement change – Agile sprint review alignment
- Automation script failure – Regular script maintenance
- Third-party API downtime – Use mock server

## **11. Roles & Responsibilities**

- QA Lead – Strategy & Planning
- QA Engineer – Test Case Execution
- Automation Engineer – Script Development
- Development Team – Defect Fixing
- Product Owner – Requirement Clarification

## **12. Defect Management Process**

- Tool: JIRA
- Severity Levels: Critical / High / Medium / Low
- Bug Lifecycle: New → Assigned → Fixed → Retest → Closed

## **13. Schedule**

- Test Planning – 2 Days
- Test Case Design – 5 Days
- Execution – 10 Days
- Regression – 5 Days
- Reporting – 2 Days